Examiner only

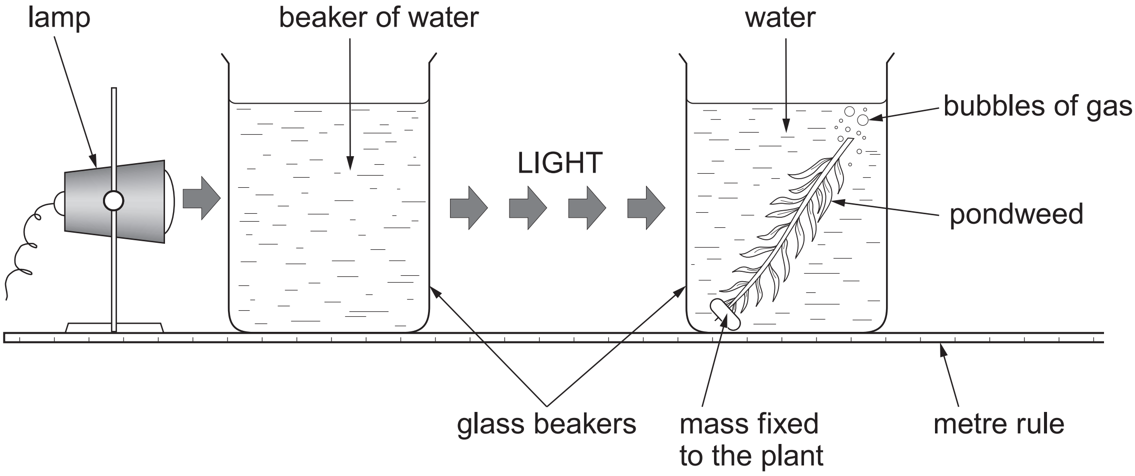
10. *(a)* (i) Write the **word** equation for photosynthesis. [2]

....

(ii) State the name of the pigment present in plant cells which absorbs light. [1]

....

Phoebe and Adam used the apparatus below to study the rate of photosynthesis in the pondweed *(Elodea sp.)*.



The number of gas bubbles per minute produced by the pondweed was counted at different distances from the light.

The experiment was carried out three times at each distance.

The results are shown below. Means were calculated to the nearest whole number.

Distance of lamp from pondweed (cm)	Number of bubbles per minute			
	Test 1	Test 2	Test 3	Mean
10	19	32	25	25
20	14	20	20	18
30	11	15	17	....
40	7	10	13	10
50	5	9	11	8

Examiner only

(b) **Complete the table** opposite by calculating the mean number of bubbles for a distance of 30 cm. **Write your answer in the table.** [2]

(c) State the relationship between the distance of the lamp from the pondweed and number of bubbles produced per minute. Explain your answer. [3]

....

....

....

....

(d) Explain why a beaker of water was placed between the lamp and the pondweed. [1]

....

....

(e) State how you could improve the accuracy of this investigation. [1]

....

10

END OF PAPER